

EXPLORATORY DATA ANALYSIS

Large-Scale Retail Transaction Dataset

Professional Analytics Project Report

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Dataset: Retail Large Dataset

Records: 100,000 **Original variables:** 18

This report presents a reproducible exploratory analysis covering data quality, descriptive statistics, distributions, relationships, outliers, correlation, multivariate patterns and business interpretation.

1. Executive Summary

The dataset contains **100,000** retail transaction records across **18 original variables**. The file contains no missing values and no exact duplicate rows. The observed overall return rate is **14.81%**, average final transaction value is **■64,166**, and average delivery time is **5.51 days**.

Home & Kitchen generates the highest total category revenue at approximately **■1,092,431,408**. Product price has a strong positive Pearson correlation with final transaction value (**0.73**). Final transaction value is right-skewed (skewness **0.93**), reflecting a smaller number of high-value transactions.

The analysis treats outliers as observations requiring investigation rather than automatic errors. All business interpretations are descriptive and should not be read as causal conclusions.

Headline Metrics

Metric	Value
Transactions	100,000
Original variables	18
Missing cells	0
Duplicate rows	0
Return rate	14.81%
Average final price	■64,166
Average discount	15.01%
Average delivery	5.51 days

2. Dataset Understanding & Data Quality

The dataset contains customer, geography, order, product, pricing, payment, shipping and return information. The original 18 fields are summarized below.

Variable	Type
customer_id	Categorical ID
age	Numeric
gender	Categorical
city	Categorical
state	Categorical
customer_segment	Categorical
order_id	Categorical ID
order_date	Date
product_category	Categorical
product_subcategory	Categorical
product_price	Numeric
quantity	Numeric
discount_percentage	Numeric
final_price	Numeric
payment_method	Categorical
shipping_type	Categorical
delivery_days	Numeric
return_status	Categorical

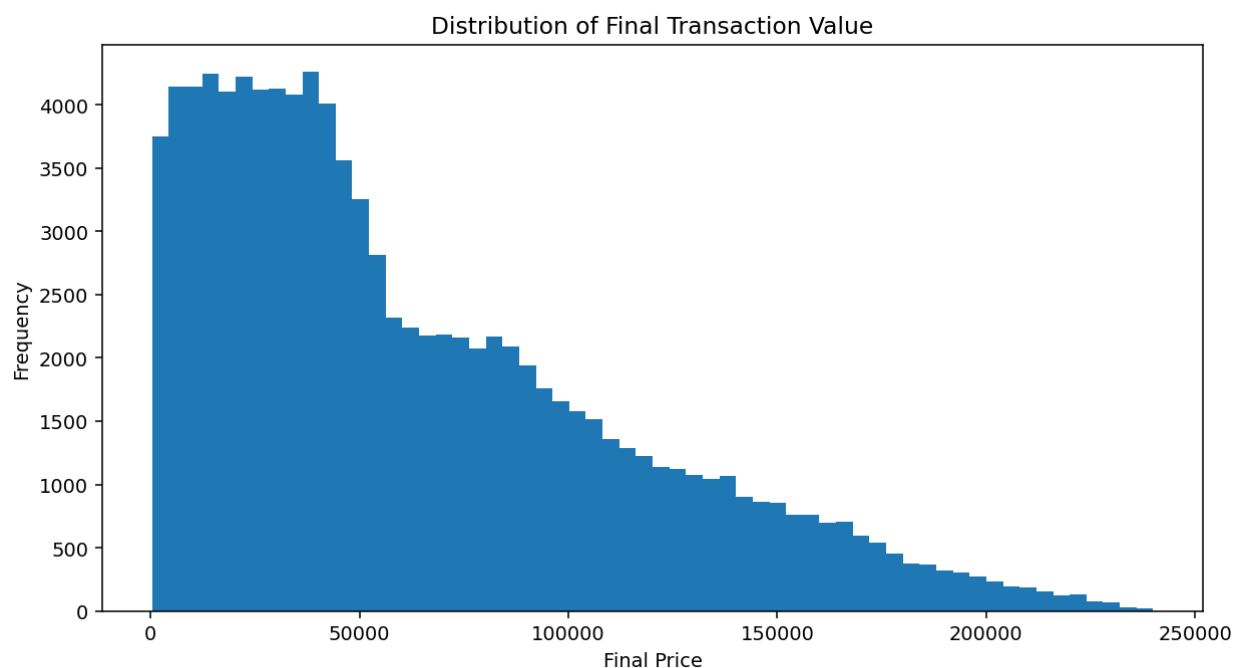
Quality Check	Result
Rows	100,000
Columns	18
Missing cells	0
Duplicate rows	0
Invalid order dates	0
Final-price formula mismatches	0

Data-quality conclusion: the uploaded dataset is structurally complete. The order-date field parses successfully, there are no missing cells, no exact duplicate records, and the final-price field agrees with $\text{product price} \times \text{quantity} \times (1 - \text{discount}\%)$ to within the validation tolerance.

3. Descriptive Statistics

The following table provides the requested descriptive statistics, including mean, median, standard deviation, minimum, quartiles, maximum, skewness and kurtosis.

Variable	count	mean	median	std	min	25%	50%	75%	max	skewness	kurtosis
age	100,000	41.56	42.00	13.85	18.00	30.00	42.00	54.00	65.00	-0.005	-1.201
product_price	100,000	30,164.53	30,148.00	17,139.68	500.00	15,375.00	30,148.00	44,957.00	59,998.00	0.003	-1.195
quantity	100,000	2.50	3.00	1.12	1.00	1.00	3.00	4.00	4.00	-0.003	-1.365
discount_percentage	100,000	15.01	15.00	8.95	0.00	7.00	15.00	23.00	30.00	-0.004	-1.202
final_price	100,000	64,165.72	49,752.39	49,969.28	357.00	24,705.00	49,752.39	94,046.07	239,980.00	0.935	0.169
delivery_days	100,000	5.51	5.00	2.87	1.00	3.00	5.00	8.00	10.00	0.002	-1.222

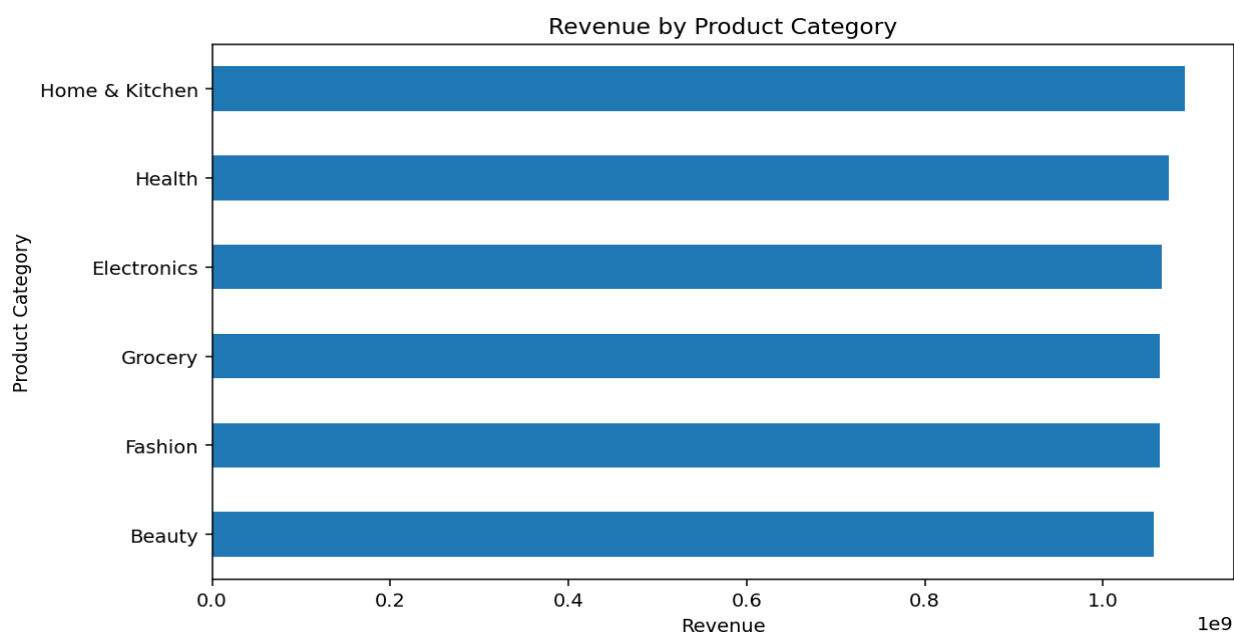


Interpretation: most operational variables are close to symmetric, while final transaction value is positively skewed (≈ 0.93). This is consistent with a distribution containing relatively more high-value observations than a symmetric distribution would contain.

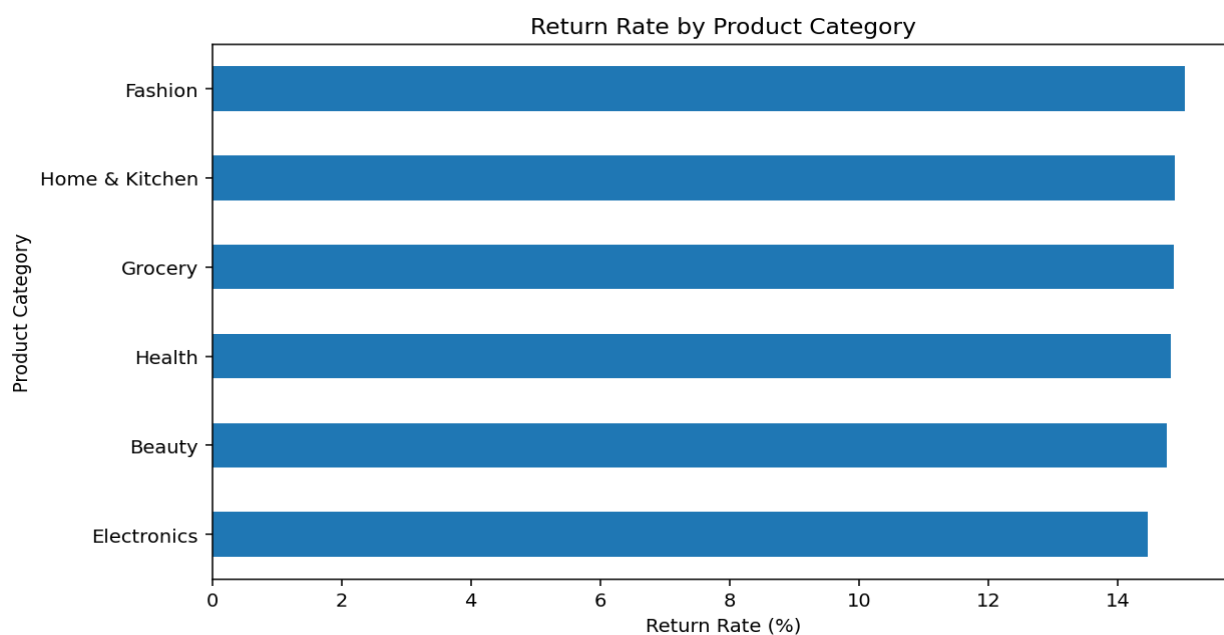
4. Univariate Analysis — Product Categories

Category performance is evaluated using transaction count, total revenue, average order value, return rate and average discount.

Product Category	Transactions	Revenue	Avg Order Value	Return Rate	Avg Discount
Home & Kitchen	16848	■1,092,431,408	■64,840	14.89%	15.04%
Health	16858	■1,073,630,252	■63,687	14.83%	14.99%
Electronics	16576	■1,066,328,623	■64,330	14.47%	14.94%
Grocery	16592	■1,063,577,206	■64,102	14.87%	15.09%
Fashion	16613	■1,063,268,353	■64,002	15.05%	14.97%
Beauty	16513	■1,057,336,495	■64,031	14.76%	15.02%



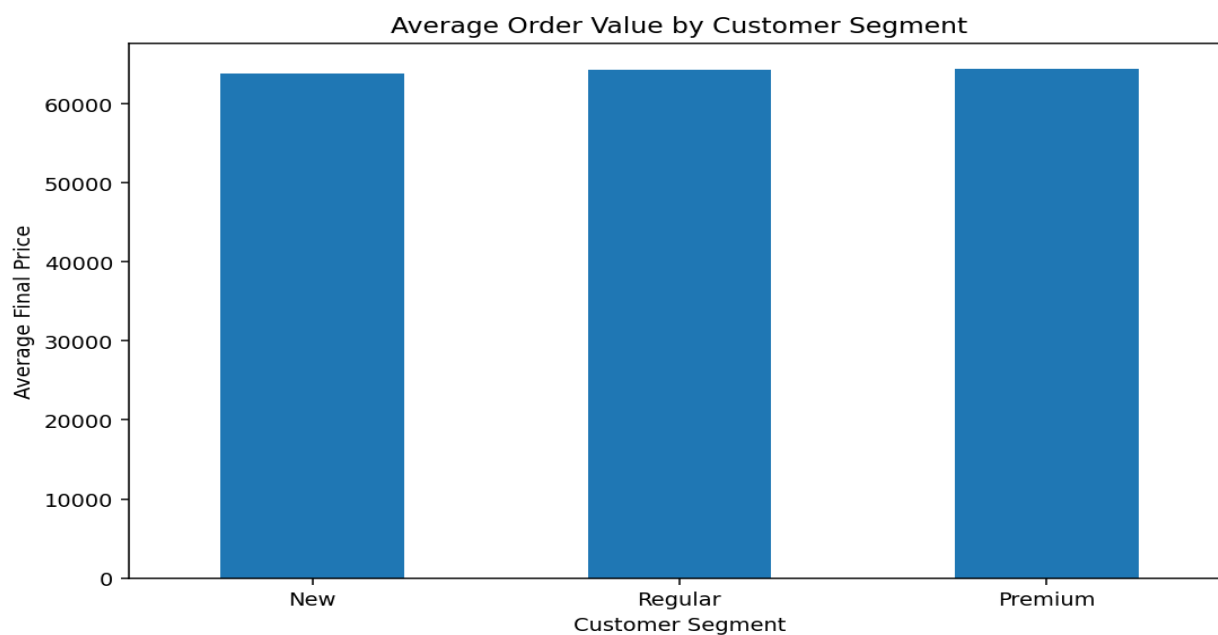
Home & Kitchen has the highest total revenue in the observed data (■1,092,431,408), while the category-level return rates are relatively close, ranging from 14.47% to 15.05%.



5. Customer Segment Analysis

Customer segments are compared using transaction volume, average and median transaction value, total revenue and return rate.

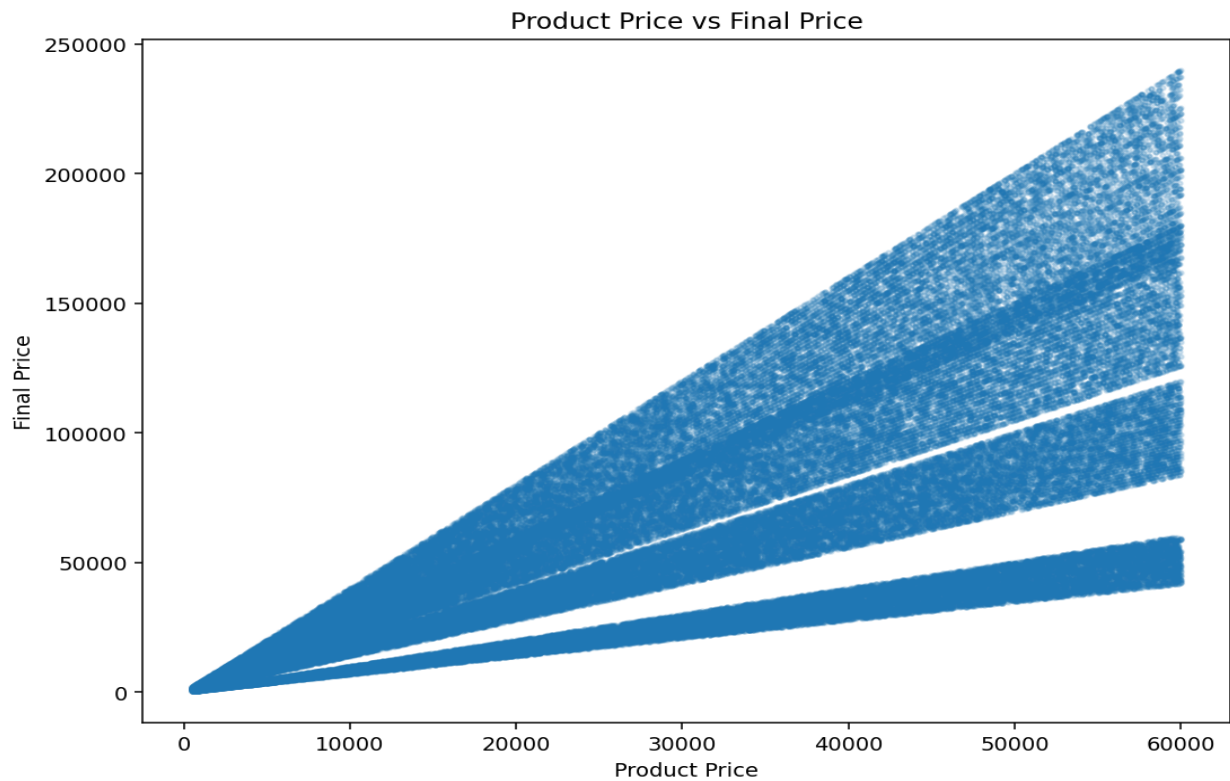
Customer Segment	Transactions	Avg Order Value	Median Order Value	Revenue	Return Rate
Premium	33151	■64,430	■50,274	■2,135,933,892	14.83%
Regular	33361	■64,303	■49,768	■2,145,217,546	14.70%
New	33488	■63,767	■49,225	■2,135,420,900	14.90%



The three segment averages are relatively close. Premium has the highest average transaction value, but the difference from Regular and New is modest compared with the overall transaction scale. Segment labels should therefore be used as descriptive categories rather than assumed evidence of substantially different spending behavior.

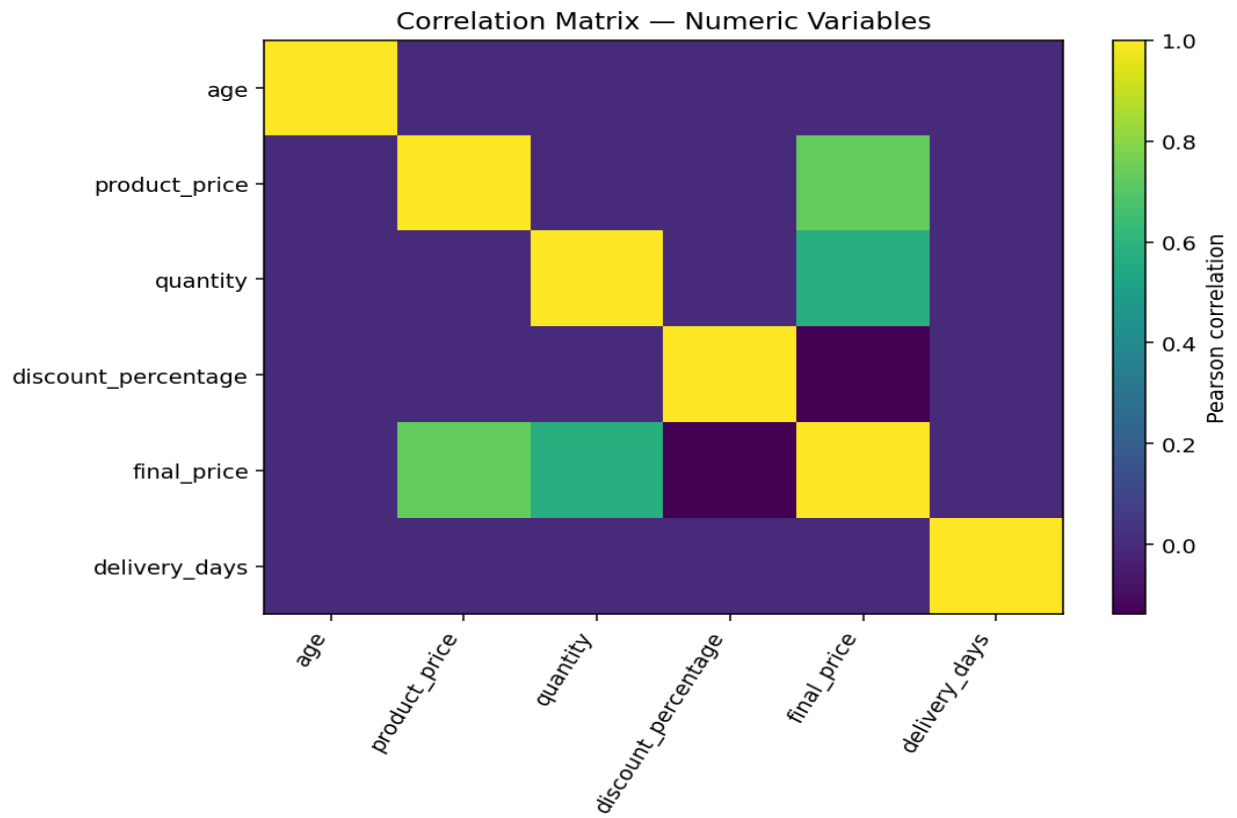
6. Bivariate & Correlation Analysis

The strongest observed numeric relationship with final transaction value is product price (Pearson $r = 0.731$). Quantity is also positively associated with final transaction value ($r = 0.575$), while discount percentage has a modest negative association ($r = -0.137$). Delivery days and age show near-zero linear association with final transaction value.



Pearson correlation matrix

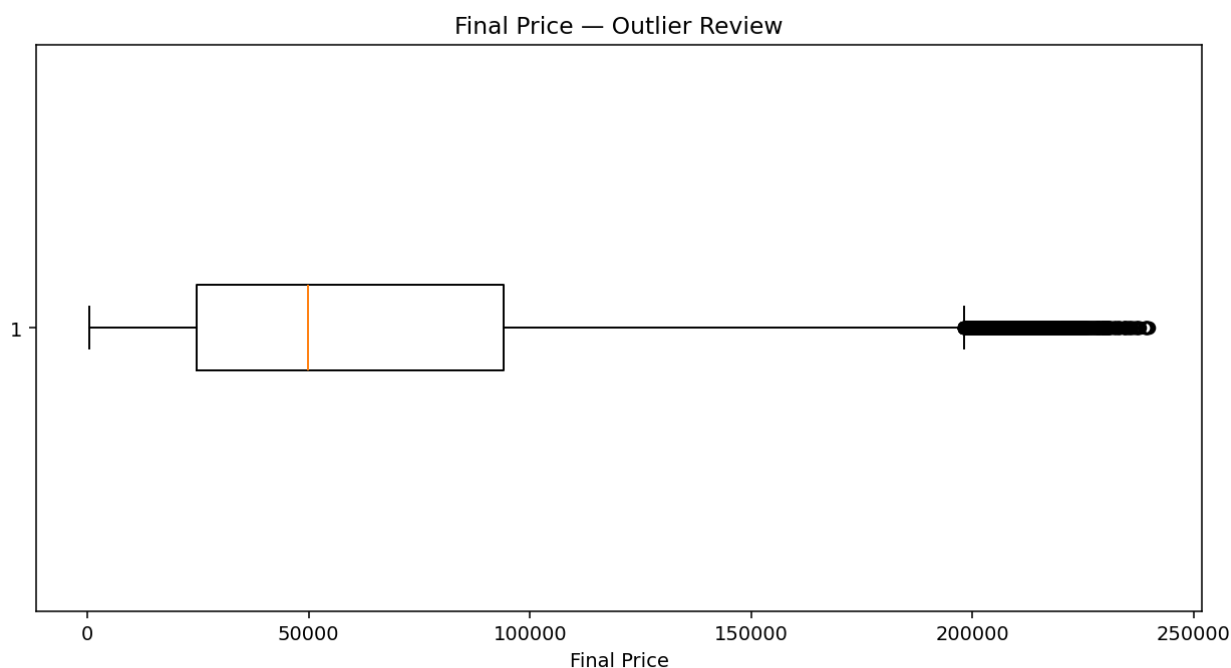
Variable	age	product_price	quantity	discount_percentage	final_price	delivery_days
age	1.0	-0.001	-0.0	-0.004	-0.001	-0.004
product_price	-0.001	1.0	0.001	-0.003	0.731	0.002
quantity	-0.0	0.001	1.0	-0.004	0.575	0.004
discount_percentage	-0.004	-0.003	-0.004	1.0	-0.137	-0.0
final_price	-0.001	0.731	0.575	-0.137	1.0	0.004
delivery_days	-0.004	0.002	0.004	-0.0	0.004	1.0



7. Outlier Analysis

Outliers were assessed using both the IQR rule ($1.5 \times \text{IQR}$ beyond the quartiles) and a standardized Z-score threshold of $|Z| > 3$.

Variable	IQR_Outliers	IQR_%	Z>3_Outliers	Z>3_%	IQR_Lower	IQR_Upper
age	0	0.00%	0	0.00%	-6.00	90.00
product_price	0	0.00%	0	0.00%	-28,998.00	89,330.00
quantity	0	0.00%	0	0.00%	-3.50	8.50
discount_percentage	0	0.00%	0	0.00%	-17.00	47.00
final_price	1335	1.33%	503	0.50%	-79,306.60	198,057.67
delivery_days	0	0.00%	0	0.00%	-4.50	15.50

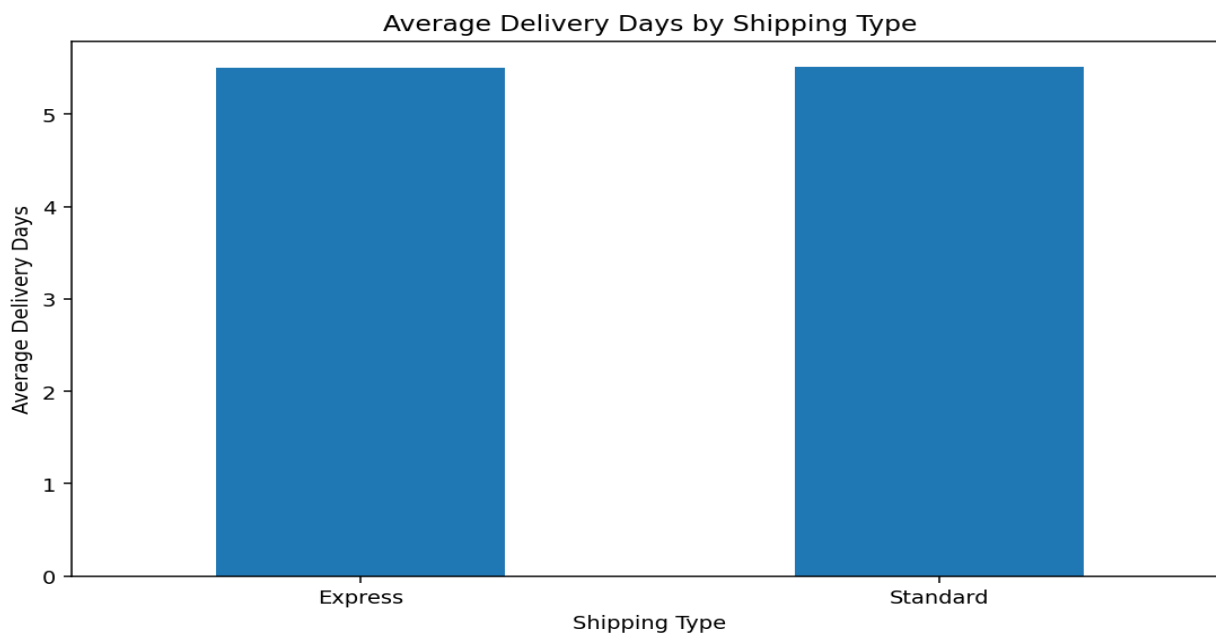


Final transaction value has 1,335 IQR-flagged observations (1.33%). Because these observations are mathematically consistent with the pricing formula, they should not be deleted solely because they are large. They may represent legitimate high-value purchases.

8. Delivery, Shipping & Returns

The overall return rate is 14.81%. Express and Standard shipping have very similar average delivery times and return rates.

Shipping Type	Transactions	Avg Delivery Days	Return Rate	Revenue
Express	50008	5.50	14.91%	■3,207,516,752
Standard	49992	5.51	14.71%	■3,209,055,586



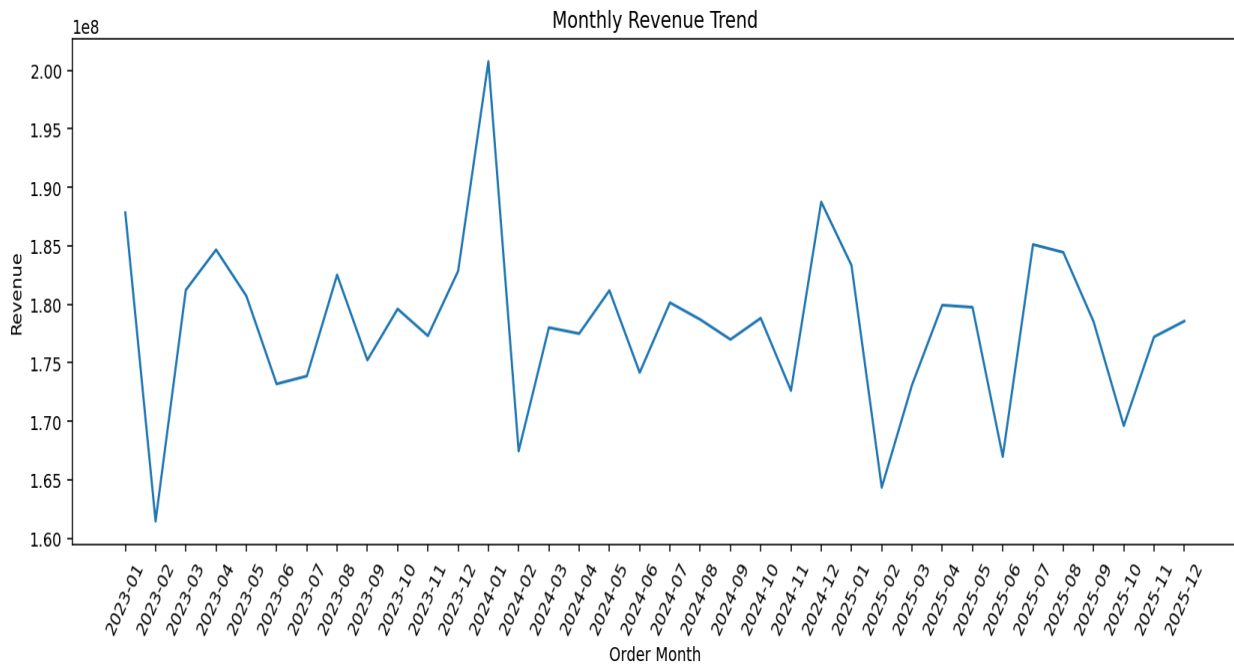
The small descriptive difference between shipping methods does not by itself establish that one shipping method causes more or fewer returns. A future analysis could control for product category, order value, city and customer segment before drawing operational conclusions.

9. Geography, Payment & Time Patterns

Maharashtra contributes the largest total revenue because it has approximately twice the transaction volume of the other individual states represented in the dataset. Tamil Nadu has the highest average transaction value among the listed states.

State	Transactions	Avg_Order_Value	Revenue
Maharashtra	19952	64308.14918855252	■1,283,076,193
Tamil Nadu	9993	65107.979726808764	■650,624,041
Telangana	10150	64081.53217438424	■650,427,552
Gujarat	10057	64102.31247787611	■644,676,957
Karnataka	10047	63795.18959191799	■640,950,270
West Bengal	9955	64286.894467101956	■639,976,034
Rajasthan	9920	64481.218115927426	■639,653,684
Uttar Pradesh	9964	63818.439802288245	■635,886,934
Delhi	9962	63370.87672254567	■631,300,674

Monthly revenue trend



The highest-revenue month in the observed period is 2024-01, with revenue of ■200,764,028. The lowest is 2023-02, with ■161,456,829. These are descriptive historical observations within the supplied dataset and should not be interpreted as forecasts.

10. Multivariate Analysis

Multivariate analysis considers several dimensions simultaneously to avoid relying on a single-variable view.



This view helps distinguish overall discount behavior from category-specific patterns. Because final price is constructed from product price, quantity and discount, the relationship is partly mechanical; the chart is therefore most useful for identifying category clusters and unusual observations rather than claiming an independent causal effect of discount percentage.

11. Key Findings

- 1. Dataset quality:** 100,000 records were analyzed with no missing cells and no exact duplicate rows.
- 2. Returns:** 14.81% of transactions have a return status of Yes.
- 3. Category revenue:** Home & Kitchen has the highest total revenue at **■1,092,431,408**.
- 4. Customer segments:** Premium has the highest average transaction value (**■64,430**), while the three segment averages remain relatively close.
- 5. Pricing relationship:** Product price and final transaction value have a strong positive Pearson correlation of 0.731.
- 6. Outliers:** 1,335 final-price observations are flagged by the IQR rule, and the pricing formula is internally consistent.
- 7. Delivery:** Express and Standard shipping average 5.50 and 5.51 days respectively, with similar observed return rates.
- 8. Distribution:** Final transaction value is right-skewed (skewness 0.93), unlike the more symmetric operational variables.

12. Business Recommendations

- Investigate high-value transactions as a separate customer/order cohort rather than removing them as errors.
- Monitor category-level return rates and subcategory-level return behavior to identify operational or product-specific issues.
- Evaluate discounts against both transaction value and return outcomes; avoid interpreting the observed correlation as causal.
- For operational decisions, enrich the analysis with delivery promises, actual-vs-promised delivery, fulfillment center and customer-level repeat behavior if those fields become available.
- If this project is extended, use hypothesis testing or regression to quantify uncertainty and control for confounding variables.

13. Limitations & Conclusion

This project is an exploratory analysis of one supplied transaction dataset. It describes observed patterns but does not establish causality. The dataset contains one row per order ID in the supplied file, so customer-level repeat-purchase behavior cannot be inferred from repeated order IDs. The absence of missing values also reflects the supplied file and does not guarantee that the underlying data-generation process is free from bias.

Overall, the dataset is clean and suitable for EDA. The analysis identifies a strong relationship between product price and final transaction value, a measurable return rate, category and segment differences, and a small population of high-value transactions that warrant business review. The accompanying notebook provides the reproducible calculations and visualizations used to generate this report.

14. Tools & Reproducibility

Python • Pandas • NumPy • Matplotlib • Seaborn • SciPy • Jupyter/Google Colab • ReportLab

Files included with this submission:

- Shailly_Maurya_Retail_EDA.ipynb — reproducible analysis notebook
- Shailly_Maurya_Retail_EDA_Report.pdf — professional final report
- retail_large_dataset_cleaned.csv — analysis-ready dataset with derived fields